Filtrado Basado en Contenido

IIC3633
Denis Parra
Profesor Asociado
PUC Chile

Contenidos

1. Contenido en lugar de ratings
2. Representación de Espacio Vectorial
3. TF-IDF
4. Buscando Items Similares
5. Representación en Espacio Latente

¿Por qué un Recomendador Basado en Contenido?

- El filtrado colaborativo tiene algunas desventajas: cold-start, sparcity, transparency.

- **PROS de content-based:**
  - A diferencia del Filtrado Colaborativo, si los items tienes descripciones suficientes, nos evitamos el "new-item problem""
  - Las representaciones del contenido son variadas y permiten utilizar diversas técnicas de procesamiento del texto, uso de información semántica, inferencias, etc.
  - Es sencillo hacer un sistema más transparente: usamos el mismo contenido para explicar las recomendaciones.

¿Por qué un Recomendador Basado en Contenido?

CONS de content-based:

- Tienden a la sobre-especialización: va a recomendar items similares a los ya consumidos, creando una tendencia al "filter bubble".

- Los métodos basados en filtrado colaborativo han mostrado ser, empíricamente, más precisos al momento de generar recomendaciones.

Arquitectura de Sistema de Recomendación CB

Los componentes principales:

(1) Analizador del Contenido,

(2) Aprendizaje del Perfil de Usuario,

(3) Filtrado de Contenido

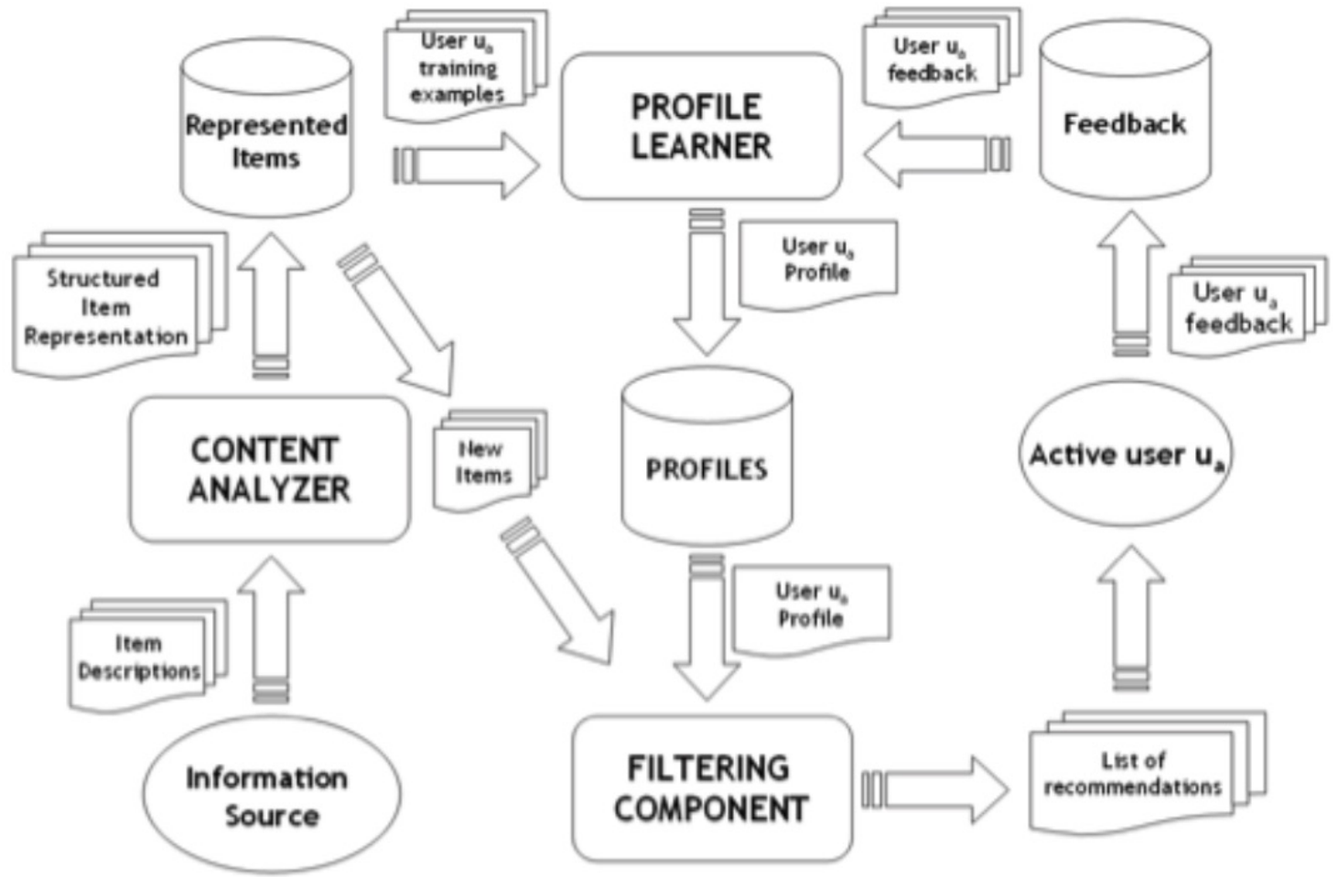
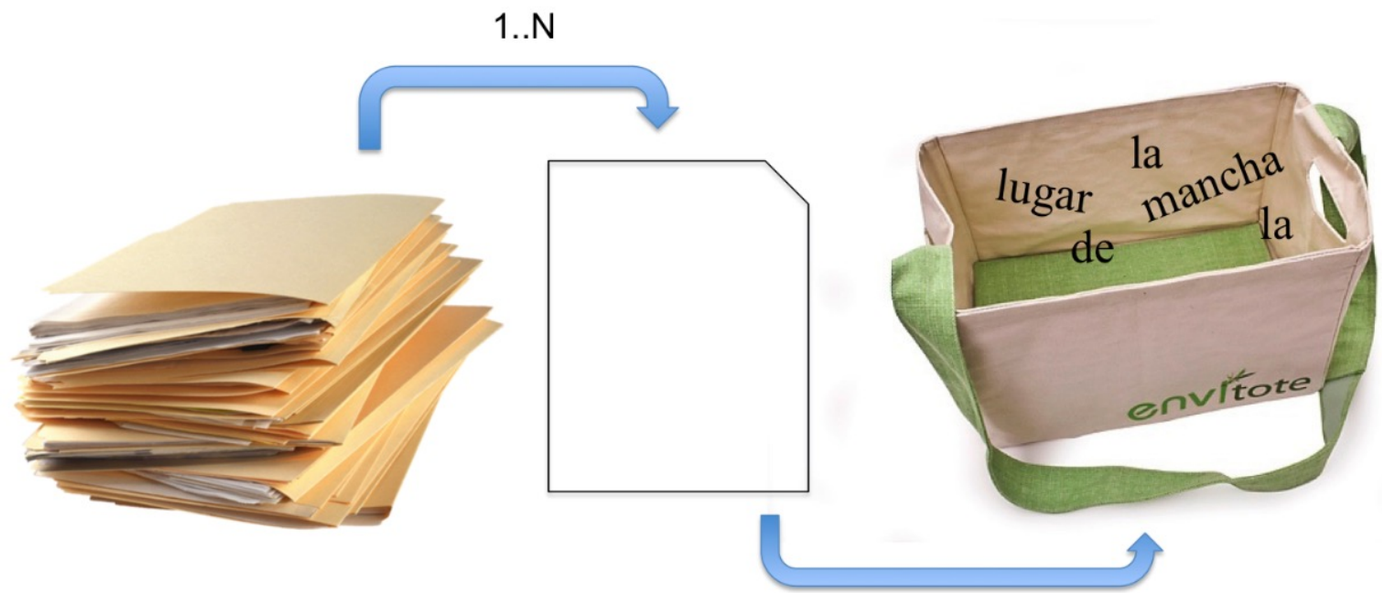


Fig. 3.1: High level architecture of a Content-based Recommender

Representación del Contenido: Bolsa de Palabras

- Se suele representar a los documentos como "bolsas de palabras"; de esta forma es fácil pasar a representar cada documento como un vector (Vector Space Model)



Representación del Contenido: VSM

- El corpus completo puede entonces representarse como una matriz donde las filas son términos y las columnas son documentos.

	Anthony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth . . .
ANTHONY	1	1	0	0	0	1
BRUTUS	1	1	0	1	0	0
CAESAR	1	1	0	1	1	1
CALPURNIA	0	1	0	0	0	0
CLEOPATRA	1	0	0	0	0	0
MERCY	1	0	1	1	1	1
WORSER	1	0	1	1	1	0
. . .						

- Luego, ¿Cúal es la mejor forma de representar los pesos de los términos?

Representación del Contenido: VSM II

Frecuencia de los términos

- Cada documento se representa como un vector, el "peso" de cada palabra para ese documento puede darse en base a la frecuencia del término en el documento.

$$\mathrm{TF}(t_k, d_j) = \frac{f_{k,j}}{max_z f_{z,j}}$$

	Anthony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth . . .
ANTHONY	157	73	0	0	0	1
BRUTUS	4	157	0	2	0	0
CAESAR	232	227	0	2	1	0
CALPURNIA	0	10	0	0	0	0
CLEOPATRA	57	0	0	0	0	0
MERCY	2	0	3	8	5	8
WORSER	2	0	1	1	1	5
. . .						

- Podemos normalizar el valor en función de la frecuencia máxima de cualquier término en el documento.

Representación del Contenido: VSM III

Log de Frecuencia de los términos

- Pero el hecho que un término aparece 100 veces y otro término sólo 10 veces, no hace al primero 10 veces más relevantes: usar logaritmo

▪La log-frecuencia del término t en d se define como

$$w_{t,d} = \begin{cases} 1 + \log_{10} \mathrm{tf}_{t,d} & \text{if } \mathrm{tf}_{t,d} > 0 \\ 0 & \text{otherwise} \end{cases}$$

▪$\mathrm{tf}_{t,d} \rightarrow w_{t,d}$:

$0 \rightarrow 0,\ \ 1 \rightarrow 1,\ \ 2 \rightarrow 1.3,\ \ 10 \rightarrow 2,\ \ 1000 \rightarrow 4,$ etc.

TF-IDF

- Bajo la intuición de que un término que aparece en sólo unos poco documentos podría ser descriptivo, podemos considerar la "Inverse Document Frequency" y combinarla con la "Term Frequency":

$$\text{TF-IDF}(t_k, d_j) = \underbrace{\text{TF}(t_k, d_j)}_{\text{TF}} \cdot \underbrace{log \frac{N}{n_k}}_{\text{IDF}}$$

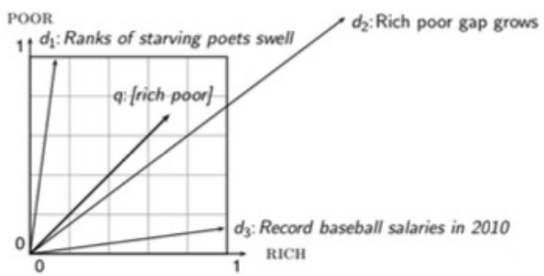
donde t_k es el término k, d_j es el documento j.

Resumen de componentes del VSM

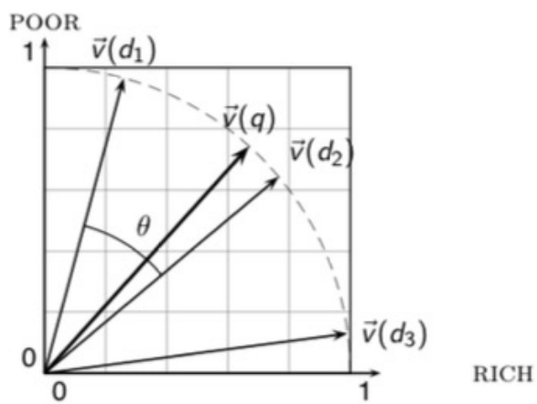
Term frequency		Document frequency		Normalization	
n (natural)	$\mathrm{tf}_{t,d}$	n (no)	1	n (none)	1
l (logarithm)	$1+\log(\mathrm{tf}_{t,d})$	t (idf)	$\log \frac{N}{\mathrm{df}_t}$	c (cosine)	$\frac{1}{\sqrt{w_1^2+w_2^2+\ldots+w_M^2}}$
a (augmented)	$0.5+\frac{0.5\times \mathrm{tf}_{t,d}}{\max_t(\mathrm{tf}_{t,d})}$	p (prob idf)	$\max\{0, \log \frac{N-\mathrm{df}_t}{\mathrm{df}_t}\}$	u (pivoted unique)	$1/u$
b (boolean)	$\begin{cases} 1 & \text{if } \mathrm{tf}_{t,d}>0 \\ 0 & \text{otherwise} \end{cases}$			b (byte size)	$1/CharLength^{\alpha}$, $\alpha<1$
L (log ave)	$\frac{1+\log(\mathrm{tf}_{t,d})}{1+\log(\mathrm{ave}_{t\in d}(\mathrm{tf}_{t,d}))}$				

Buscando ítems similares

Distancia Euclidiana



Distancia Coseno



$$sim(d_i, d_j) = \frac{\sum_k w_{ki} \cdot w_{kj}}{\sqrt{\sum_k {w_{ki}}^2} \cdot \sqrt{\sum_k {w_{kj}}^2}}$$

Buscando ítems similares II

- Okapi BM25

$$RSV_d = \sum_{t \in q} IDF \cdot \frac{(k_1 + 1) tf_{td}}{k_1((1-b) + b \times (L_d / L_{ave})) + tf_{td}} \cdot \frac{(k_3 + 1) tf_{tq}}{k_3 + tf_{tq}}$$

Denis Parra and Peter Brusilovsky. 2009.
Collaborative filtering for social tagging systems: an experiment with CiteULike.
In Proceedings of the third ACM conference on Recommendersystems (RecSys '09) http://doi.acm.org/10.1145/1639714.1639757

Pre-procesamiento del texto

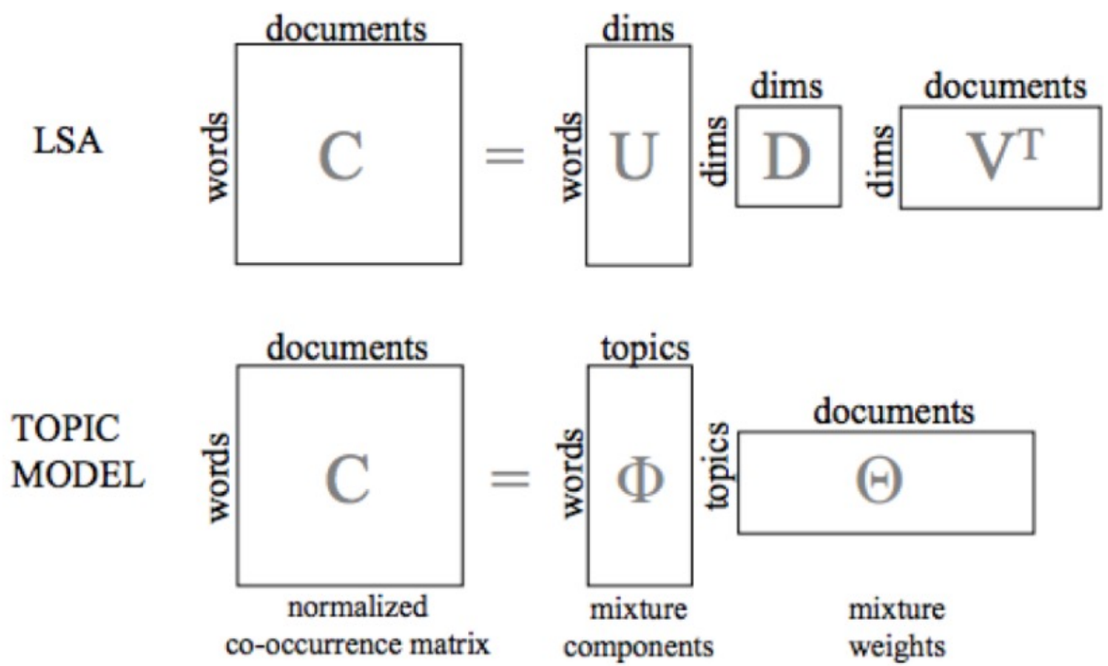
- Pasar a mayúsculas/minúsculas
- Tokenization
- Stemming (Porter, Krovetz)
- Lemmatization
- Etc.

Representación Semántica del Contenido

- No todo el contenido del documento corresponde a la misma categoría.
- Autor, palabras clave, fechas, tópicos pueden dar una noción adicional de filtrado.
- Opción 1: Representación semántica explícita (No lo veremos en detalle en esta clase)
  - Ontologías
  - WordNet
  - ConceptNet
- Opción 2: Inferir representación semántica (LSI, LDA)
- Opción 3: Word Vectors (Word2Vec, Glove)

Representación de documentos en espacio latente: LSA y LDA

- Latent Semantic Indexing/Analysis
- Latent Dirichlet Allocation



LSI

	d_1	d_2	d_3	d_4	d_5	d_6
ship	1	0	1	0	0	0
boat	0	1	0	0	0	0
ocean	1	1	0	0	0	0
voyage	1	0	0	1	1	0
trip	0	0	0	1	0	1

=

	1	2	3	4	5
ship	−0.44	−0.30	0.57	0.58	0.25
boat	−0.13	−0.33	−0.59	0.00	0.73
ocean	−0.48	−0.51	−0.37	0.00	−0.61
voyage	−0.70	0.35	0.15	−0.58	0.16
trip	−0.26	0.65	−0.41	0.58	−0.09

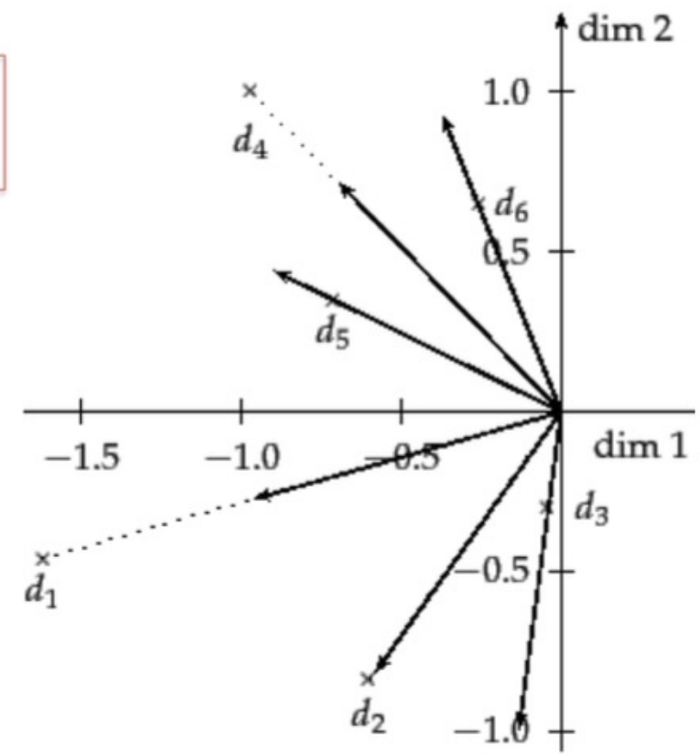
$$\begin{matrix} 2.16 & 0.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 1.59 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 1.28 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 1.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 0.00 & 0.39 \end{matrix}$$

	d_1	d_2	d_3	d_4	d_5	d_6
1	−0.75	−0.28	−0.20	−0.45	−0.33	−0.12
2	−0.29	−0.53	−0.19	0.63	0.22	0.41
3	0.28	−0.75	0.45	−0.20	0.12	−0.33
4	0.00	0.00	0.58	0.00	−0.58	0.58
5	−0.53	0.29	0.63	0.19	0.41	−0.22

M = U S V

LSI II

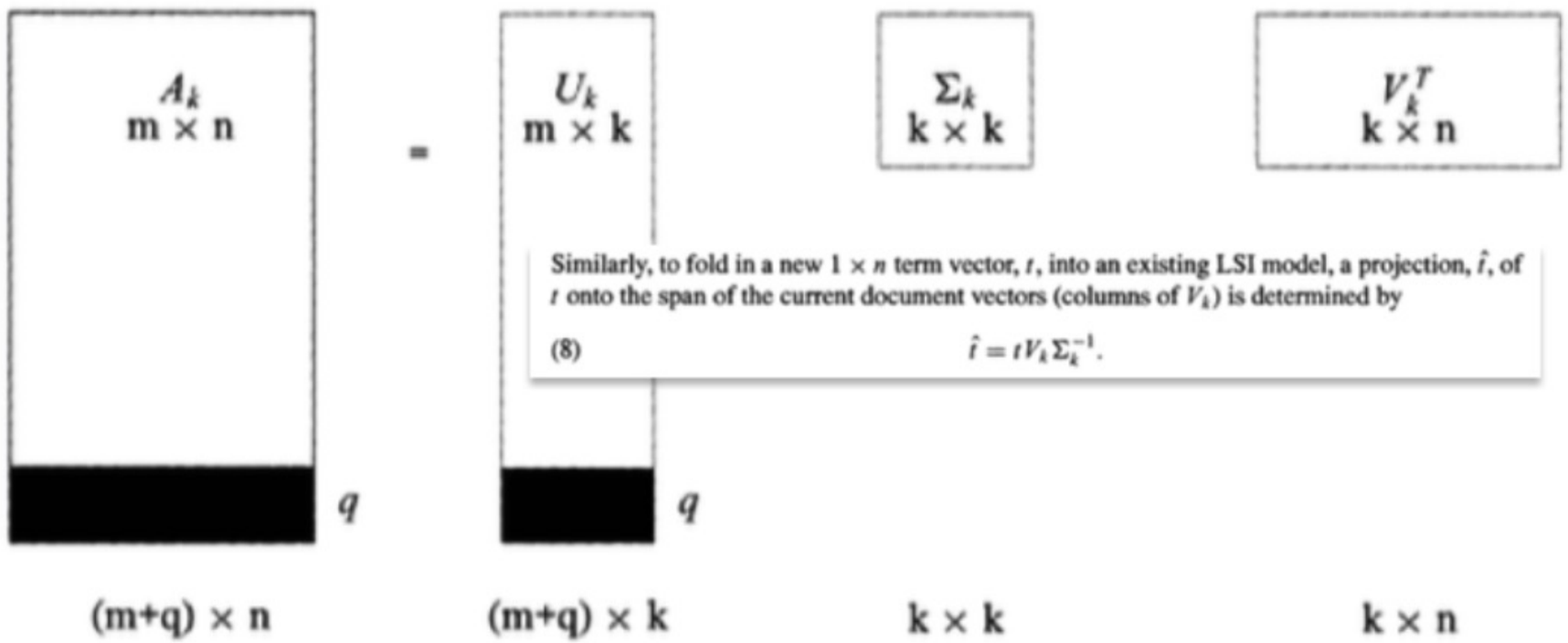
	d_1	d_2	d_3	d_4	d_5	d_6
1	−0.75	−0.28	−0.20	−0.45	−0.33	−0.12
2	−0.29	−0.53	−0.19	0.63	0.22	0.41
3	0.28	−0.75	0.45	−0.20	0.12	−0.33
4	0.00	0.00	0.58	0.00	−0.58	0.58
5	−0.53	0.29	0.63	0.19	0.41	−0.22



LSI II : http://dfao-uc.github.io

Proyección de documentos o términos nuevos

- **Folding in**: Using Linear Algebra for Intelligent Information Retrieval

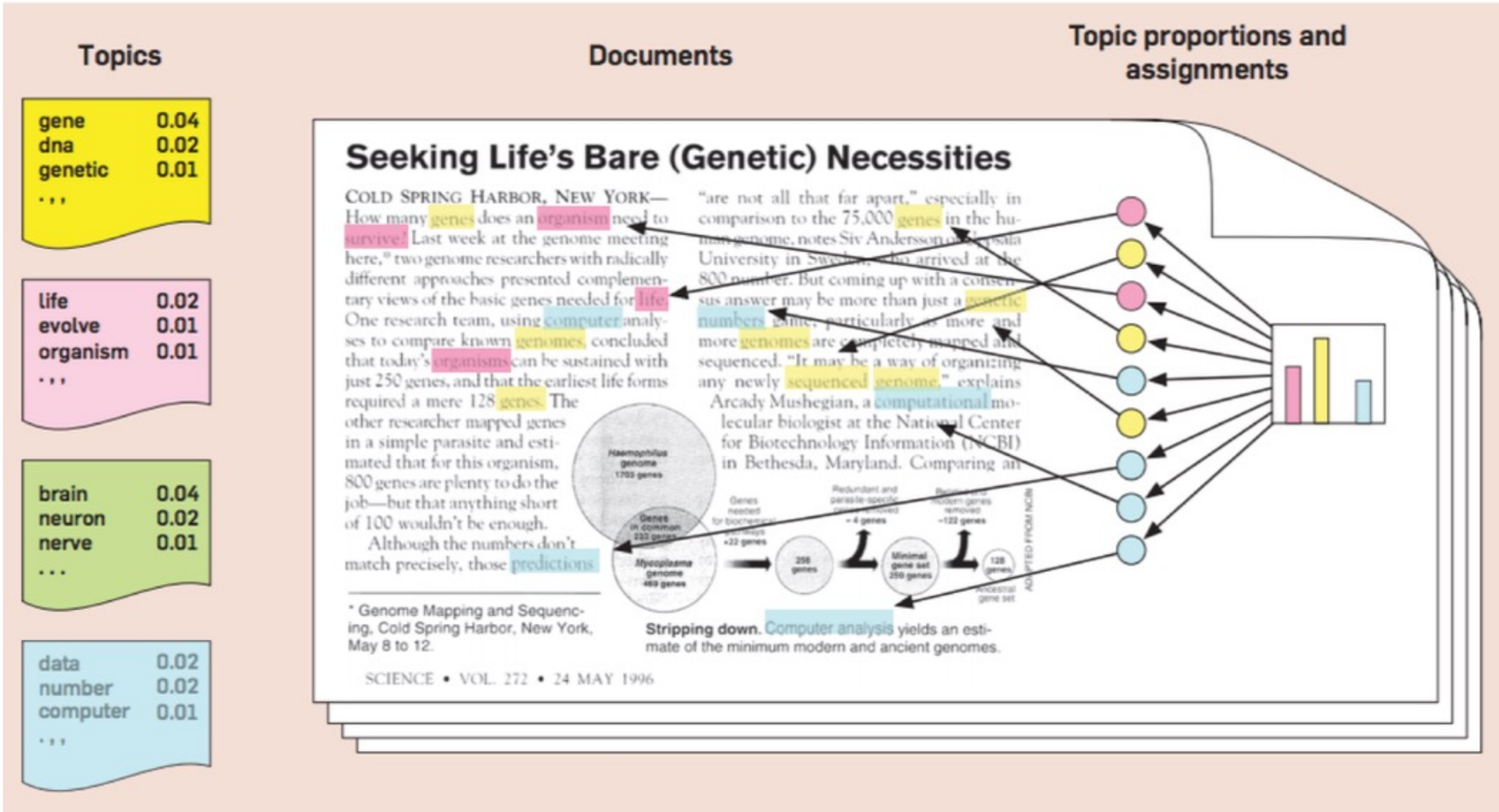


Similarly, to fold in a new $1 \times n$ term vector, t, into an existing LSI model, a projection, $\hat{t}$, of t onto the span of the current document vectors (columns of V_k) is determined by

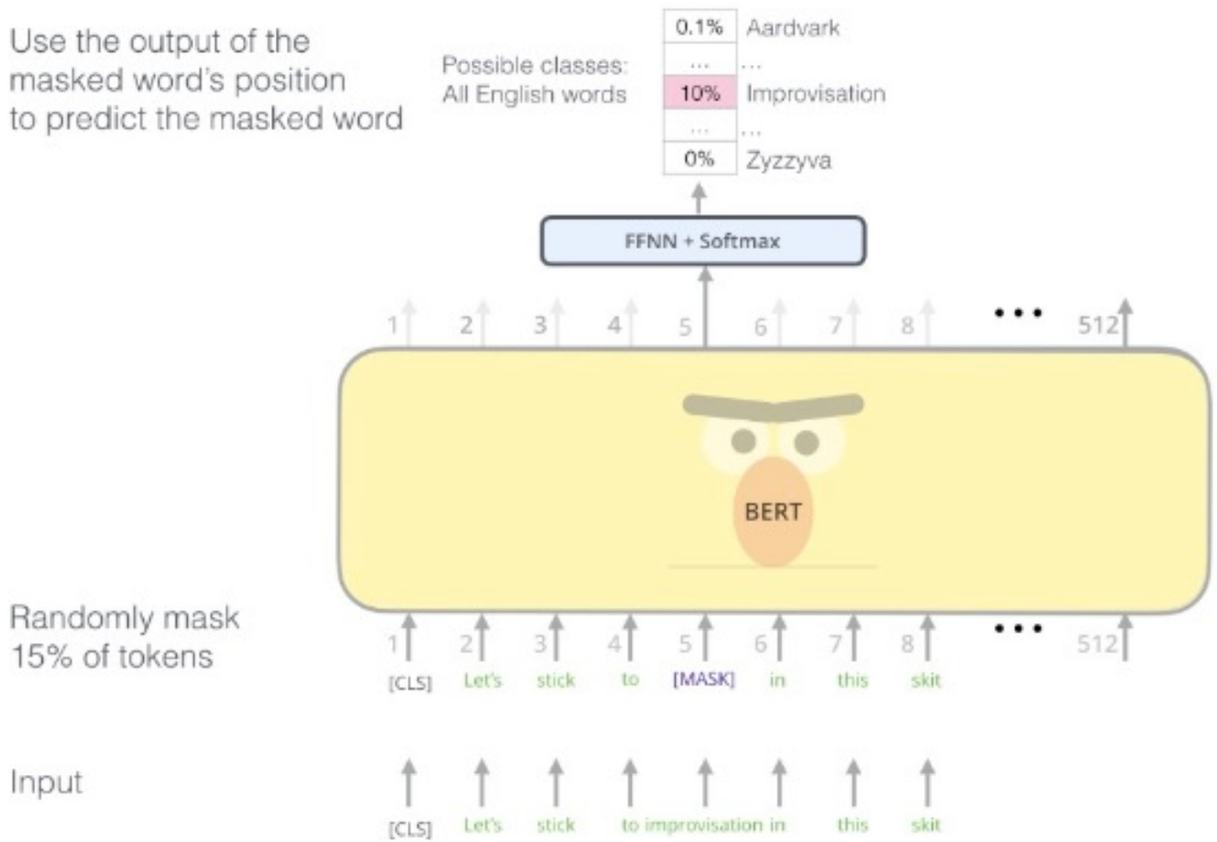
$$\hat{t} = t V_k \Sigma_k^{-1}. \tag{8}$$

FIG. 3. *Mathematical representation of folding-in q terms.*

LDA


Topics
gene 0.04
dna 0.02
genetic 0.01
...
life 0.02
evolve 0.01
organism 0.01
...
brain 0.04
neuron 0.02
nerve 0.01
...
data 0.02
number 0.02
computer 0.01
...
Documents
Topic proportions and assignments
Seeking Life's Bare (Genetic) Necessities
COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.
Although the numbers don't match precisely, those predictions
"are not all that far apart," especially in comparison to the 75,000 genes in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a genetic numbers game, particularly as more and more genomes are completely mapped and sequenced. "It may be a way of organizing any newly sequenced genome," explains Arcady Mushegian, a computational molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an
* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.
Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.
SCIENCE • VOL. 272 • 24 MAY 1996

Codificando texto con NNs: BERT



BERT's clever language modeling task masks 15% of words in the input and asks the model to predict the missing word.

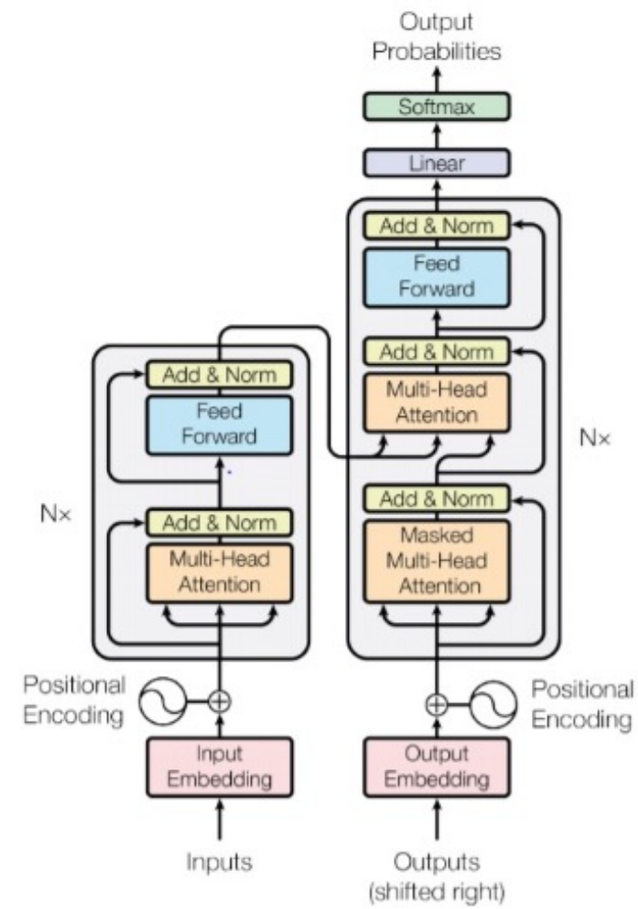
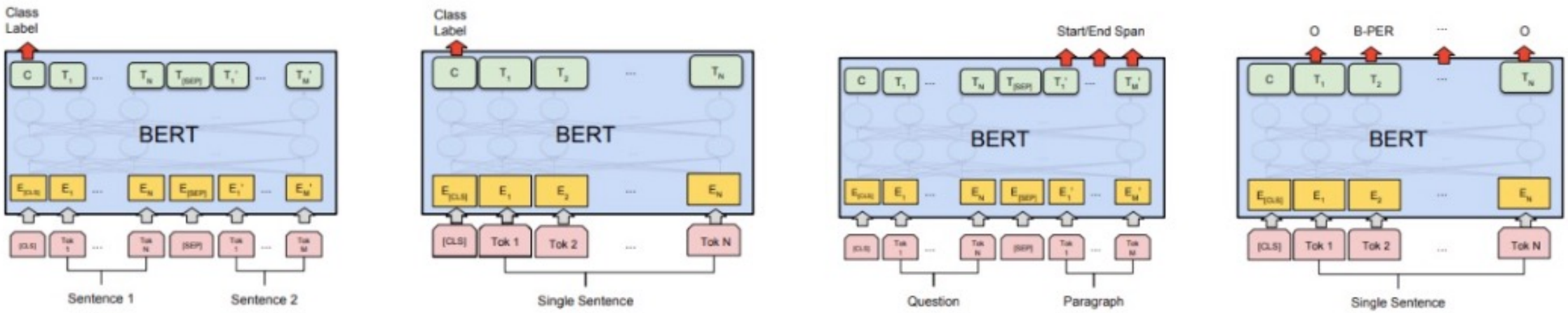


Figure 1: The Transformer - model architecture.

BERT para diferentes tareas



(a) Sentence Pair Classification Tasks: MNLI, QQP, QNLI, STS-B, MRPC, RTE, SWAG

(b) Single Sentence Classification Tasks: SST-2, CoLA

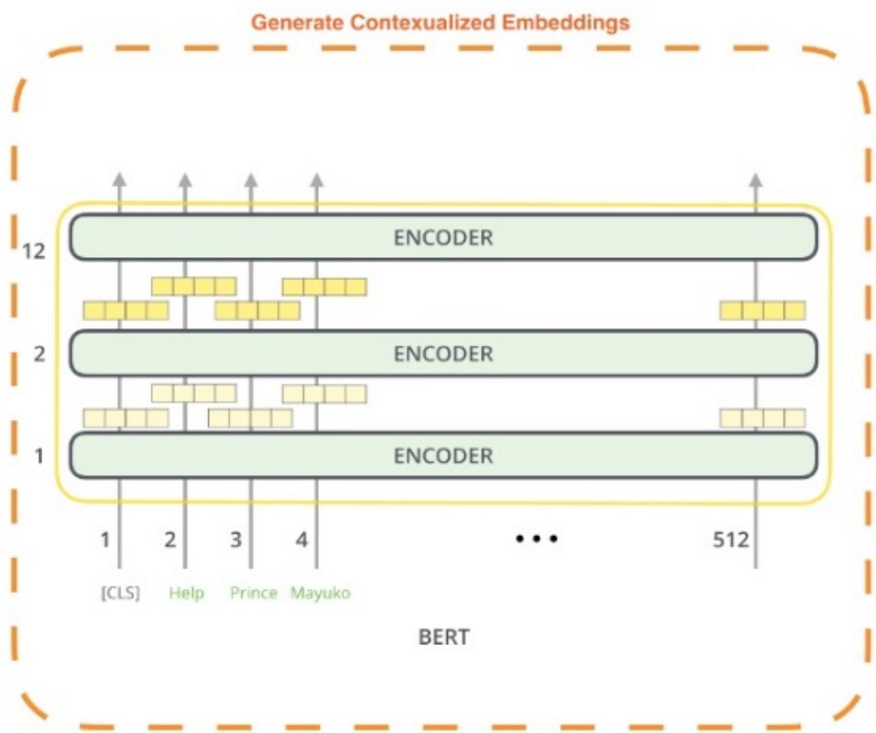
(c) Question Answering Tasks: SQuAD v1.1

(d) Single Sentence Tagging Tasks: CoNLL-2003 NER

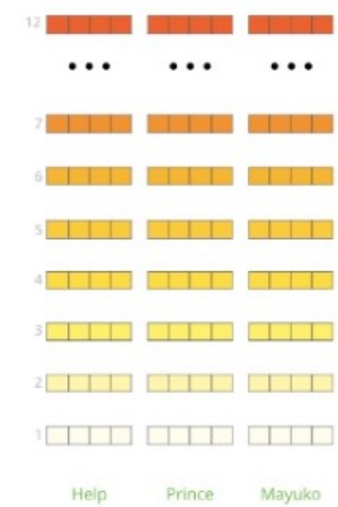
BERT para extraer características

BERT for feature extraction

The fine-tuning approach isn't the only way to use BERT. Just like ELMo, you can use the pre-trained BERT to create contextualized word embeddings. Then you can feed these embeddings to your existing model – a process the paper shows yield results not far behind fine-tuning BERT on a task such as named-entity recognition.



The output of each encoder layer along each token's path can be used as a feature representing that token.



But which one should we use?

Neural Networks for text-aware content-based recommendation

Content-Based Citation Recommendation

Chandra Bhagavatula
Allen Institute for AI
chandrab@allenai.org

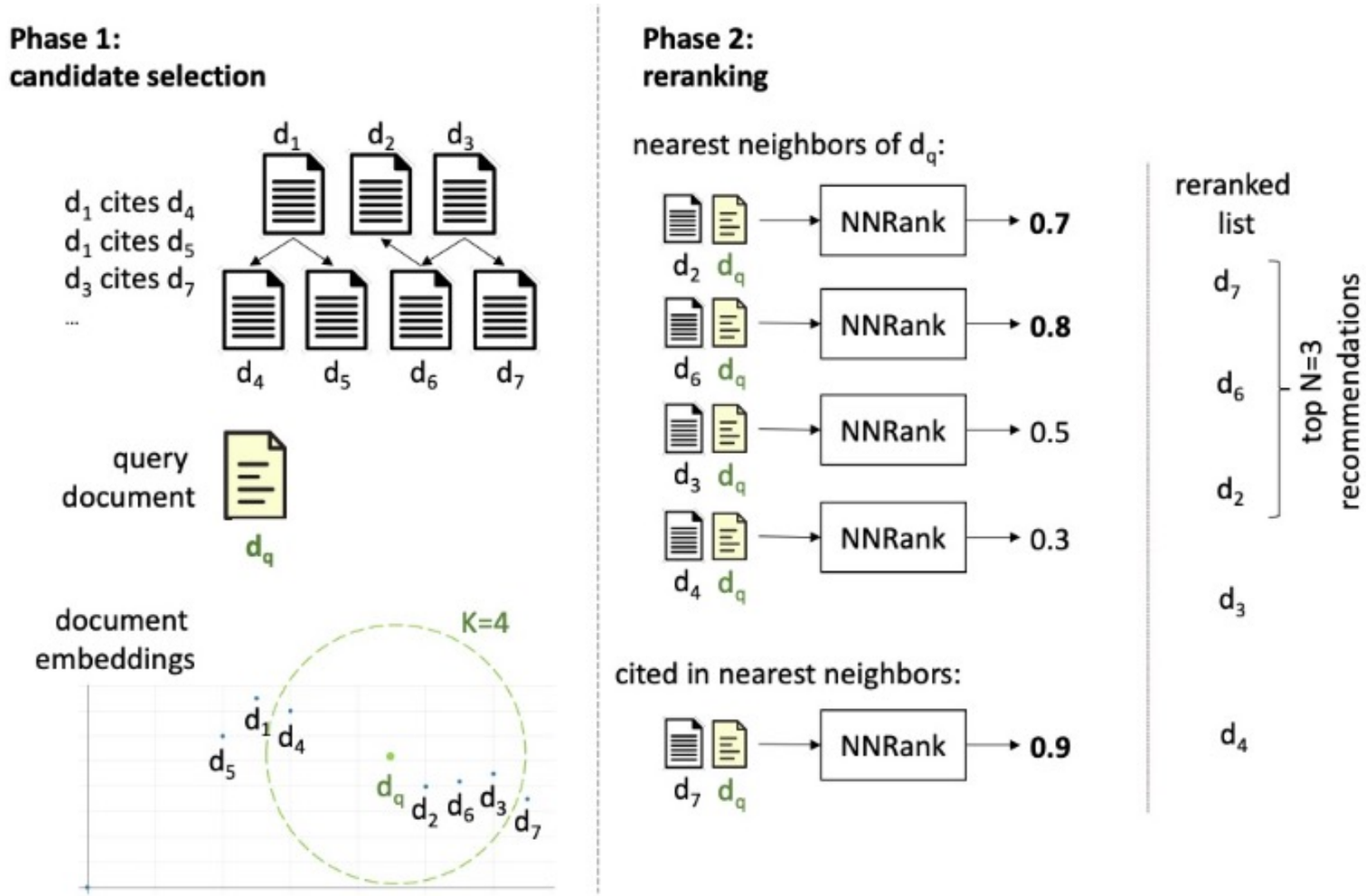
Sergey Feldman
Data Cowboys *
sergey@data-cowboys.com

Russell Power
Independent Researcher †
russell.power@gmail.com

Waleed Ammar
Allen Institute for AI
waleeda@allenai.org

https://aclanthology.org/N18-1022.pdf

2-stage model: filter and re-ranking



NNRank of stage 2

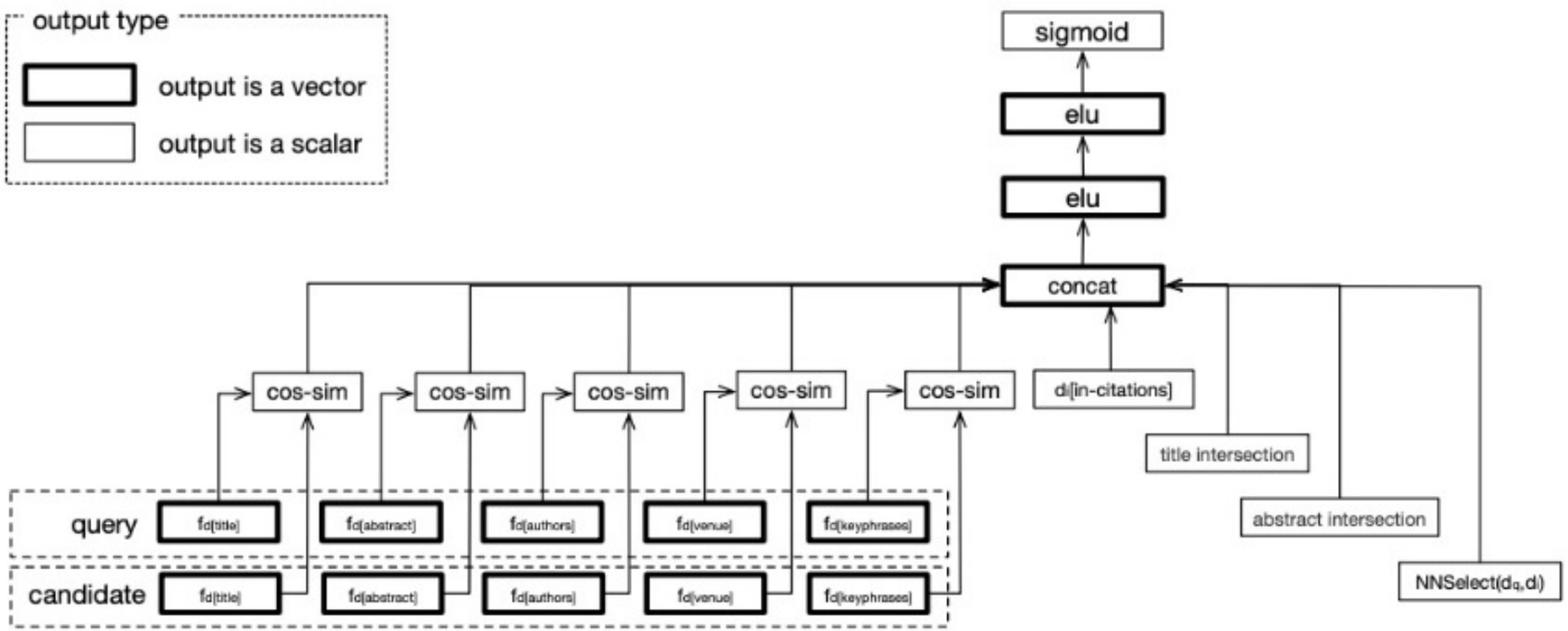


Figure 2: `NNRank` architecture. For each of the textual and categorical fields, we compute the cosine similarity between the embedding for d_q and the corresponding embedding for d_i. Then, we concatenate the cosine similarity scores, the numeric features and the summed weights of the intersection words, followed by two dense layers with ELU non-linearities The output layer is a dense layer with sigmoid non-linearity, which estimates the probability that d_q cites d_i.

Resultados

Method	DBLP		PubMed		OpenCorpus	
	F1@20	MRR	F1@20	MRR	F1@20	MRR
BM25	0.119	0.425	0.209	0.574	0.058	0.218
ClusCite	0.237	0.548	0.274	0.578	–	–
NNSelect	0.282±0.002	0.579±0.007	0.309±0.001	0.699±0.001	0.109	0.221
+ NNRank	0.302±0.001	0.672±0.015	0.325±0.001	0.754±0.003	**0.126**	0.330
+ metadata	**0.303**±0.001	**0.689**±0.011	**0.329**±0.001	**0.771**±0.003	0.125	**0.330**

Table 2: F1@20 and MRR results for two baselines and three variants of our method. BM25 results are based on our implementation of this baseline, while ClusCite results are based on the results reported in Ren et al. (2014). "NNSelect" ranks candidates using cosine similarity between the query and candidate documents in the embedding space (phase 1). "NNSelect + NNRank" uses the discriminative reranking model to rerank candidates (phase 2), without encoding any of the metadata features. "+ metadata" encodes the metadata features (i.e., keyphrases, venues and authors), achieving the best results on all datasets. Mean and standard deviations are reported based on five trials.

News recommendation

UNBERT: User-News Matching BERT for News Recommendation

Qi Zhang, **Jingjie Li***, **Qinglin Jia**, **Chuyuan Wang**, **Jieming Zhu**, **Zhaowei Wang** and **Xiuqiang He**

Huawei Noah's Ark Lab

{zhangqi193, lijingjie1, jiaqinglin2, wangchuyuan, jamie.zhu, wangzhaowei3, hexiuqiang1}@huawei.com

https://www.ijcai.org/proceedings/2021/0462.pdf

Objetivo de UNBERT

- Separar la contribución a nivel de palabra y a nivel de noticia

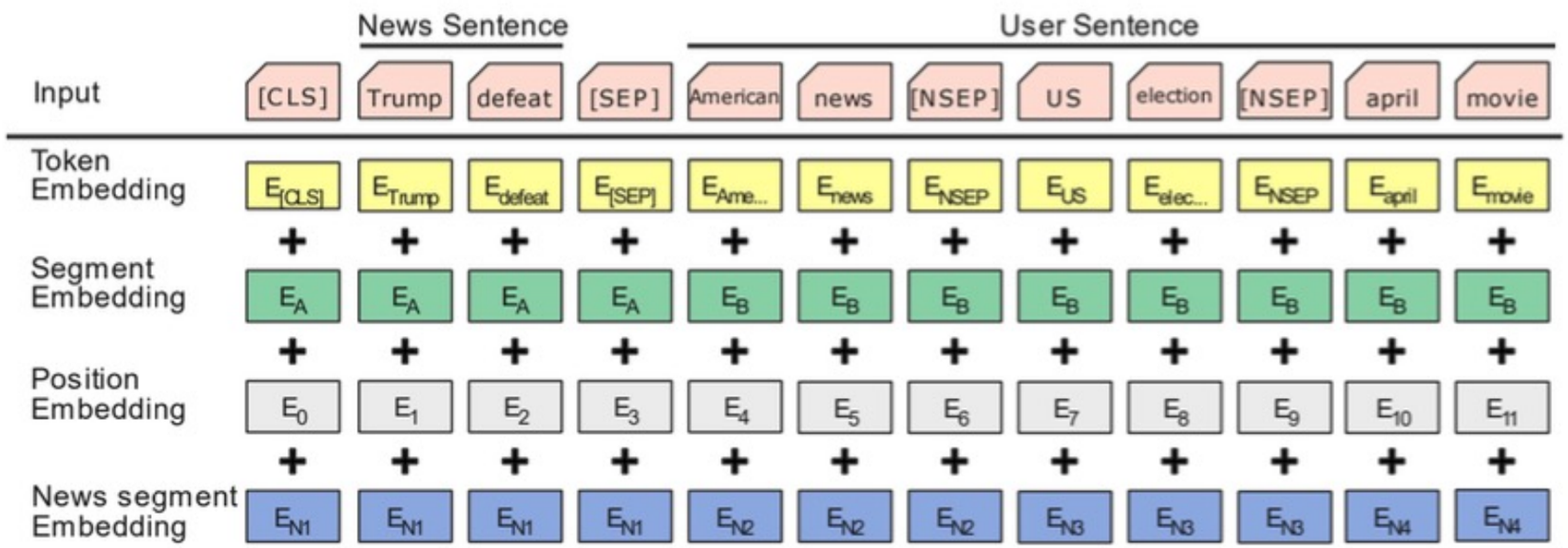


Figure 2: UNBERT input representation. The input embeddings are the sum of the token embeddings, the segmentation embeddings, the position embeddings, and the News segmentation embeddings.

Arquitectura de UNBERT

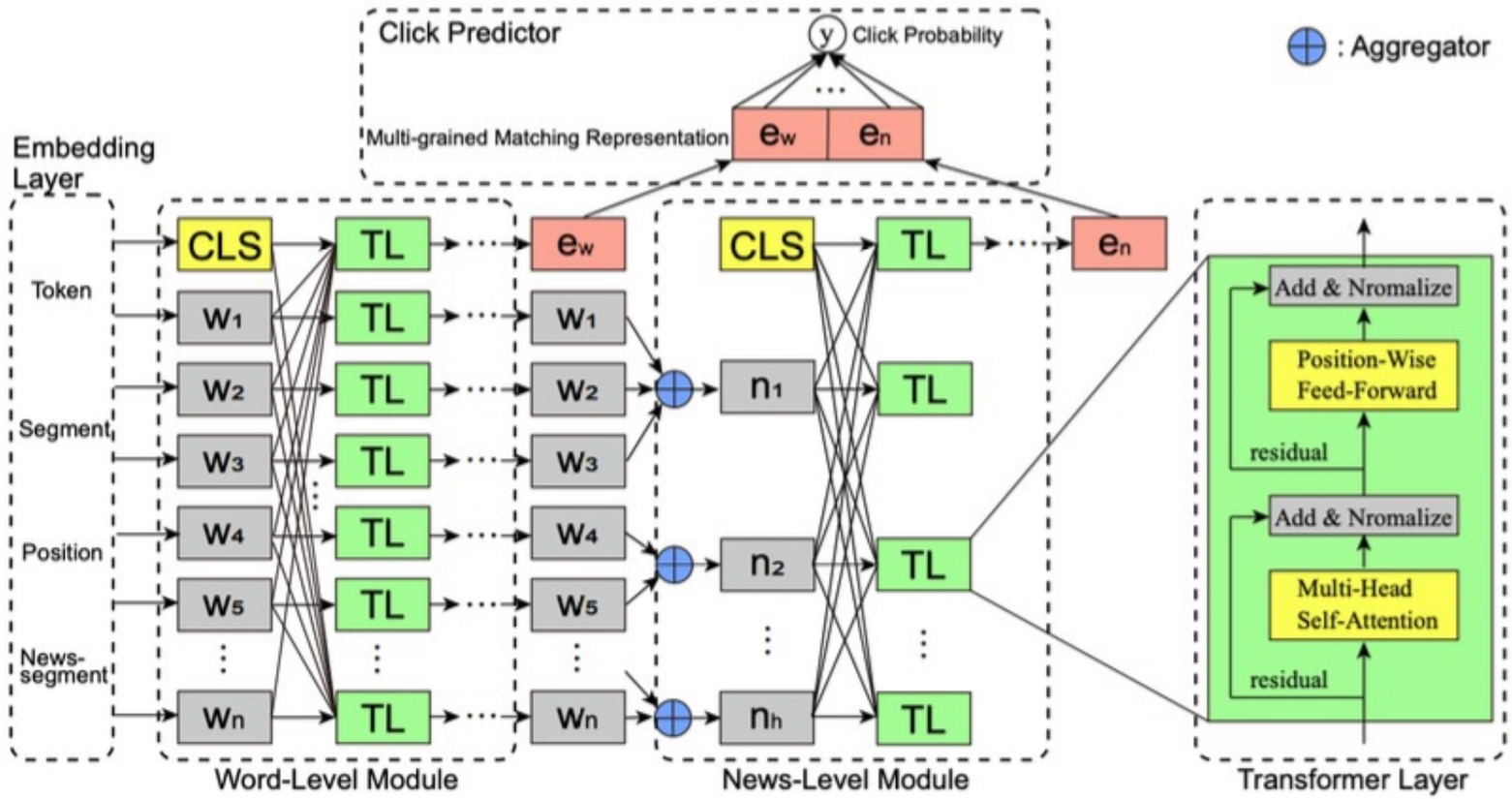


Figure 3: The overall architecture of our UNBERT approach.

Resultados de UNBERT

Method	MIND-small				MIND-large			
	AUC	MRR	nDCG@5	nDCG@10	AUC	MRR	nDCG@5	nDCG@10
LibFM	0.5974	0.2633	0.2795	0.3429	0.6185	0.2945	0.3145	0.3713
DeepFM	0.5989	0.2621	0.2774	0.3406	0.6187	0.2930	0.3135	0.3705
DKN	0.6175	0.2705	0.2890	0.3538	0.6407	0.3042	0.3292	0.3866
NPA	0.6321	0.2911	0.3170	0.3781	0.6592	0.3207	0.3472	0.4037
NAML	0.6550	0.3039	0.3308	0.3931	0.6646	0.3275	0.3566	0.4140
LSTUR	0.6438	0.2946	0.3189	0.3817	0.6708	0.3236	0.3515	0.4093
NRMS	0.6483	0.3001	0.3252	0.3892	0.6766	0.3325	0.3628	0.4198
FIM	0.6502	0.3026	0.3291	0.3910	0.6787	0.3346	0.3653	0.4221
UNBERT	**0.6762**	**0.3172**	**0.3475**	**0.4102**	**0.7068**	**0.3568**	**0.3913**	**0.4478**
%Improv.	2.12	1.33	1.67	1.71	2.81	2.22	2.60	2.57
UNBERT-en$^{\triangle}$	-	-	-	-	0.7183	0.3659	0.4020	0.4581

Boldface indicates the best results (the higher, the better), while the second best is underlined. UNBERT-en$^{\triangle}$ represents the ensemble score based on UNBERT which is at the top of https://msnews.github.io/#leaderboard.

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